Department of Physics, Princeton University

Graduate Preliminary Examination
Part I

Thursday, May 1, 2003
9:00 am - 12:00 noon

Answer TWO out of the THREE questions in Section A (Mechanics) and TWO out of the THREE questions in Section B (Electricity and Magnetism).

Work each problem in a separate examination booklet. Be sure to label each booklet with your name, the section name, and the problem number.

Section A. Mechanics

1. The earth is in a circular orbit of angular frequency ω about the sun. The sun is so much more massive than the earth that, for our purposes, it may be taken to sit at rest at the center of our coordinate system. Lagrange discovered that there exist a certain number of equilibrium points at which an artificial satellite of negligible mass can orbit the sun with the same frequency ω as the earth (while maintaining a fixed distance from both the earth and the sun). Such orbits are ideally suited for space-based observatories of various kinds. We will explore some of the properties of the 'Lagrange points' in this problem.

   (a) Consider points on the line that runs from the sun through the earth. This line is of course stationary in the reference frame that rotates with the earth in its orbit. Show that there is one point on this line outside the earth's orbit where a test particle may sit at equilibrium in the rotating frame. This point is commonly designated as the L2 Lagrange point. (There is a similar Lagrange point, L1, inside the earth's orbit as well.)

   (b) Give an approximate expression, correct to leading order in the small quantity $\beta = M_e/M_s$, for the distance from the earth to the L2 Lagrange point described above. Express your answer in terms of the masses and R, the earth-sun distance. The Wilkinson Microwave Anisotropy Probe (WMAP) is stationed at L2: using $\beta \approx 3 \times 10^{-6}$ and $R = 1.5 \times 10^8$ km, find the distance from earth to WMAP.

   (c) Determine whether the L2 equilibrium point is stable or unstable against small perturbations in position along the earth-sun line.

2. The problem of a sphere rolling without slipping directly down an inclined plane is an old chestnut of freshman physics. In this problem, you are asked to examine more general motions of a sphere rolling without slipping on inclined and rotating planes.

   The state of motion of a sphere of mass M is specified by giving its position $\vec{R}$ and velocity $d\vec{R}/dt$ with respect to a fixed point and its angular velocity $\vec{\omega}$ with respect to its center of mass. The rolling without slipping constraint is usually expressed as $d\vec{R}/dt = a\vec{\omega} \times \hat{n}$ where a is the sphere's radius and $\hat{n}$ is a unit vector normal to the plane on which the sphere is rolling. There is a constraint force $\vec{f}$ which acts in the plane to enforce the constraint and this constraint force has to be taken into account in the equations for acceleration of the center of mass and for time rate of change of the angular momentum $I\vec{\omega}$ about the center of mass.

   (a) Suppose that the plane is inclined to the horizontal at an angle θ in the earth's gravitational field. Show that you can eliminate the constraint force to find an equation for the center of mass alone and show that the center of mass experiences an acceleration of $\frac{5}{7}g\sin\theta$ along the 'downhill' direction in the plane. Hence, the trajectories of this rolling sphere are parabolae.

   (b) Now consider the case that the plane is not inclined, but is rotating with angular velocity $\vec{\Omega}$ about the vertical axis. The most important modification to the calculation you have just done is that the condition for rolling without slipping changes. Use the changed condition in your previous analysis to show that the center of mass executes circular motion in the horizontal *inertial* frame with frequency $\frac{2}{7}\Omega$! In other words, the rolling sphere executes Larmor orbits, just like a charged particle in a uniform magnetic field!

   For reference, note that the moment of inertia of a homogeneous sphere is $\frac{2}{5}Ma^2$.

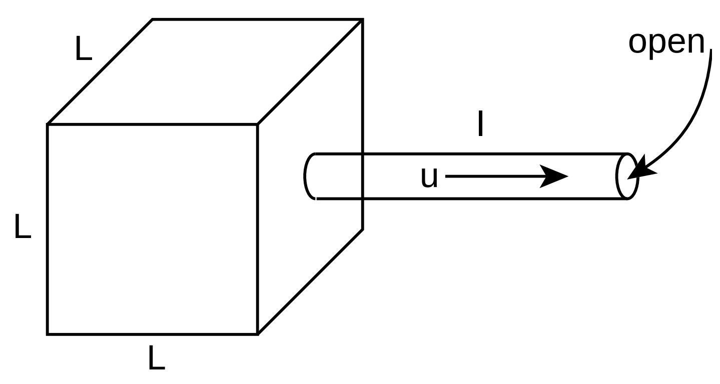
3. Consider a cube of size L filled with air. Air in the cube has an equilibrium pressure p_0 and density ρ_0.

   (a) Write the wave equation for small fluctuations $p_1(\vec{x}, t)$ of the pressure field. State the relation between fluctuations ρ_1 in the density and fluctuations p_1 in the pressure (it involves the velocity of sound c_s). What is the equation that would allow one to infer the local velocity $\vec{v}$ of the air fluid from the gradient of the fluctuating pressure field?

   (b) State the proper boundary condition which must be satisfied by the pressure field at the walls of the cube and use it to derive an expression for the eigenfrequencies of oscillation of the air in the cube. Explain why the mode which has no spatial dependence ($p = p(t)$) and eigenfrequency zero is not actually in the spectrum. Note that all real modes have frequencies that scale with the size of the cube as $1/L$.

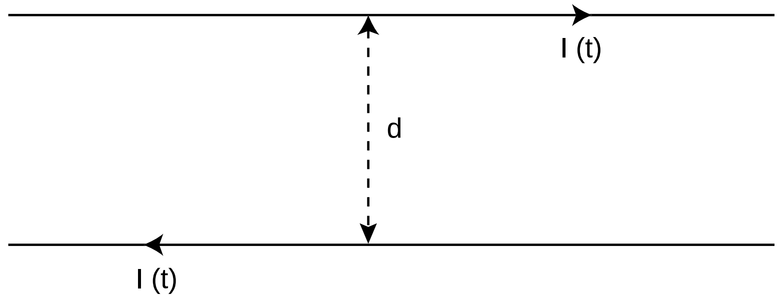
   (c) Now make a small hole of area S in a wall of the cube and attach an *open* tube of the same area and length l so that air can flow in and out of the cube. Revisit the problem of a mode in which the pressure is independent of position. A pressure excess inside the cube will accelerate the air in the tube, leading to an outflow of air which will then reduce the pressure inside the cube. Write down the coupled equations of motion for the pressure $p(t)$ in the cube and the speed $u(t)$ of airflow in the tube. Solve to find the oscillation frequency of this 'zero mode' and note that it scales as $1/L^{3/2}$ (if l and S are held fixed). This device for making low-frequency sound is called a Helmholtz resonator.

The setup is as indicated in the following figure:



Section B. Electricity and Magnetism

1. Two infinitely long parallel wires, a distance d apart, carry time-dependent currents $I(t)$ of the same magnitude but opposite direction, as shown in the figure:



We will consider two possible time histories of the current:

(a) Suppose that the current switches on suddenly at time $t = 0$ and remains constant thereafter (i.e. that $I(t) = I_0\theta(t)$). Calculate, as a function of time, the force per unit length $F(t)$ on the wires and sketch your result. It is not really possible to instantaneously turn on a current in a circuit: explain how your answer reflects this fact.

(b) Now let's do it in a more realistic fashion by turning on a linearly increasing current at time $t = 0$ which continues to increase until the value I_0 is reached (i.e. $I(t) = bt$ for $0 < t < I_0/b$ and $I = I_0$ for later times). As in the previous item, calculate the force per unit length $F(t)$ between the wires and plot your result.

Note the following possibly useful indefinite integral:

$$\int \frac{dy}{\sqrt{y^2 - a^2}} = \log(\sqrt{y^2 - a^2} + y)$$

2. The 'method of images' allows us to solve many problems involving point charges and spherical conductors of various kinds. In this question you are asked to derive and then use this method in some representative applications

   (a) Consider a charge Q placed at a distance $R > a$ from the center of a sphere of radius a (for the moment, this is just a geometrical sphere, not a conductor or any other physical object). Show that if a certain charge of opposite sign is placed inside the sphere at the appropriate location then the spherical surface of radius a is a $V = 0$ equipotential surface. This result can be used in the following applications:

   (b) A point charge Q is placed at a distance $R > a$ from the center of a conducting sphere of radius a. Find the force exerted on the sphere if the total charge on the sphere is Q.

   (c) Consider the distribution of the total charge Q on the surface of the sphere under the conditions just described. Find an equation for the distance R such that surface charge density of opposite sign to Q first appears somewhere on the sphere.

   (d) Two perfectly conducting spheres of radius a are placed far apart (their centers are separated by $R \gg 2a$) and kept at the same potential V_0 (this condition could be enforced by connecting the spheres with a fine wire). What is the charge on each sphere, correct to first order in a/R?

   (e) What is the charge on the two spheres correct to second order in a/R?

3. The E-vector of a plane electromagnetic wave propagating along the z-axis and having a polarization vector $\vec{e} = (\alpha_x, \alpha_y, 0)$ can be written

$$\vec{E}(\vec{r}, t) = E(\alpha_x \hat{x} + \alpha_y \hat{y})\; e^{i(kz - \omega t)}$$

(the polarization vector is taken to be of unit length, $\vec{\alpha} \cdot \vec{\alpha}^* = 1$). In free space, the dispersion relation is $\omega = ck$ and the wave propagates with both phase and group velocity equal to c.

Now let the wave propagate through a dilute plasma containing a density N of free mobile electrons of mass m and charge e (along with a background of compensating positive charge taken to be so massive as to be fixed in place). By solving for the motion of an electron in the electric field of the propagating wave (i.e. ignoring the effect of the wave's B-field on the motion) one can infer a polarization density $\vec{P} \propto \vec{E}$. This in turn allows us to infer a frequency-dependent dielectric constant via

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P} = \epsilon(\omega) \vec{E}\ .$$

The index of refraction of the plasma is then given by $n(\omega) = \sqrt{\epsilon/\epsilon_0}$.

(a) Use this line of argument to compute the frequency dependence of the index of refraction $n(\omega)$ of a plasma. Turn your result into a dispersion relation $\omega(k)$ and find the limiting frequency ω_p as the wavelength of the wave goes to infinity. This cutoff frequency, below which waves cannot propagate, is called the plasma frequency.

(b) Now let the plasma be subject to a static magnetic field $B_0 \hat{z}$ in the direction of propagation of the wave. Also assume that the electrons have some kinetic energy so that, in the absence of any other pertubation, they execute circular motion about the static magnetic field lines at the Larmor frequency $\omega_L = |eB_0|/m$. Extend your calculation of the dielectric constant of the plasma to this new case. The response is different for different states of circular polarization so, for definiteness, analyse the case of right circular polarization of the propagating wave ($\vec{\alpha} = (\hat{x} + i\hat{y})/\sqrt{2}$).

(c) Find the lowest frequency at which such a wave can propagate. Express your answer in terms of ω_p and ω_L.

Department of Physics, Princeton University

Graduate Preliminary Examination
Part II

Friday, May 2, 2003
9:00 am - 12:00 noon

Answer TWO out of the THREE questions in Section A (Quantum Mechanics) and TWO out of the THREE questions in Section B (Thermodynamics and Statistical Mechanics).

Work each problem in a separate booklet. Be sure to label each booklet with your name, the section name, and the problem number.

Section A. Quantum Mechanics

1. A particle with spin one-half is at rest in a static magnetic field of strength $B_z = B_0$ oriented along the z-axis. The magnetic moment interaction splits the $S_z = +1/2$ from the $S_z = -1/2$ state One can manipulate the spin state of the particle by subjecting it to a time-dependent field $\vec{B}_1 = B_1(cos\phi(t), sin\phi(t), 0)$ that rotates in the $x-y$ plane at a variable frequency $\dot{\phi}(t)$. We will discuss different ways of choosing $\phi(t)$ to achieve the goal of transforming an initial $S_z = -1/2$ state into an $S_z = +1/2$ state.

   The two-component spin wave function ψ of this system evolves under a time-dependent Hamiltonian which can be written as

$$H = \mu B_0 \sigma_z + \mu B_1(\sigma_x cos\phi(t) + \sigma_y sin\phi(t))$$

   where μ is the magnetic moment and $\sigma_{x,y,z}$ are the Pauli matrices (with $\sigma_x^2 = 1$ etc.). For general $\phi(t)$ this is hard to solve, but various special cases and approximations are helpful as we now show.

   (a) First show that the 'interaction picture' wave function

$$\hat{\psi} = \exp(-i\phi(t)\frac{\sigma_z}{2})\psi$$

   evolves according to a simpler Hamiltonian $H_{rot}(t)$ which becomes time-independent when ϕ is linear in t.

   (b) Consider the case that $\dot{\phi} = \omega_1$ for a finite time interval $-T < t < T$ and is zero otherwise (this is not too hard to realize experimentally). H_{rot} is now time-independent and you can solve the Schrödinger equation. Find ω_1 and T such that a spin-down state is perfectly converted into a spin-up state by the $-T \to T$ time evolution (this is sometimes called a 'Π pulse'). Note that for this method to work for a collection of spins, they must all be subject to the same field $B_0\hat{z}$.

   (c) Now consider the case of a 'chirped' frequency such that $\dot{\phi}(t) = \alpha t$ for $-T < t < T$. $H_{rot}(t)$ now varies with time, but if its matrix elements vary slowly (i.e if α is small), and there is no level crossing, the adiabatic theorem should apply. This means that the system remains in the 'same' eigenstate of the instantaneous Hamiltonian for all time. Make a rough plot of the eigenenergies of the instantaneous Hamiltonian H_{rot} as a function of time for this case. Show that the lowest energy eigenstate evolves from spin down at $t = -T$ to spin up at $t = +T$ if we take $\alpha T \gg \mu B_0, \mu B_1$. Note that this method of spin flipping is insensitive to the value of B_0 and could work for a collection of spins in an inhomogeneous environment.

2. Consider the familiar problem of s-wave ($l = 0$) scattering of a particle of mass m from an attractive square-well potential of depth V_0 and radius r_0: $V = -V_0\theta(r_0 - r)$.

   (a) First, let's look at whether the potential has any s-wave bound states. Show that there is a critical potential strength V_{crit} such that for $0 < V_0 < V_{crit}$ there is *no* s-wave bound state. Put another way, show that a bound state (of zero energy) first appears at V_{crit}.

   (b) Set up an equation for determining the phase shift $\delta_0(k)$ and show that it implies that $\delta_0 \sim Ak$ as $k \to 0$. Evaluate the coefficient A as a function of V_0.

   (c) Show that A vanishes, as it should, as $V_0 \to 0$. More alarmingly, show that A blows up as the potential strength V_0 approaches V_{crit} from below!

   (d) Calculate the contribution of the s-wave phase shift to the total cross section in the limit of small k. How does the zero-energy cross-section behave as $V_0 \to V_{crit}$? Comment.

3. The total angular momentum of a valence electron in an atom of orbital angular momentum l can be either $J = l + 1/2$ or $J = l - 1/2$. The two total angular momentum states are typically split by spin-orbit interactions, leaving $2J + 1$ degenerate states of the same J, but different J_z. Upon applying a uniform magnetic field $B\hat{z}$, these magnetic substates are split by the interaction between the magnetic moment and the applied magnetic field. The perturbation Hamiltonian is

$$H_{int} = -\frac{e\hbar}{2m_e c} B(L_z + 2S_z) = -\frac{e\hbar}{2m_e c} B(J_z + S_z)$$

where the relative factor of two between L_z and S_z is due to the famous fact that the g-factor of the electron is two. The Zeeman splitting of the degenerate multiplet of total angular momentum J is thus

$$\delta E_{J,J_z} = -\frac{e\hbar B}{2m_e c} \langle J, J_z|(J_z + S_z)|J, J_z\rangle = -\frac{e\hbar B}{2m_e c}(J_z + \langle J, J_z|S_z|J, J_z\rangle)$$

To evaluate this explicitly, we need to solve the slightly nontrivial problem of evaluating the matrix elements of S_z.

(a) Consider the special case $l = 1$. Construct the $J = 3/2, 1/2$ states by angular momentum addition and evaluate $\langle J, J_z|S_z|J, J_z\rangle$ for all the states. Show that the energy splittings satisfy $\delta E_{J,J_z} = g_J J_z$ and state the two values of g_J that you have just computed.

(b) Now consider the case of general l. It is possible, but tedious, to directly verify that $\delta E = g_J^l J_z$ for the two possible J multiplets. Assuming that this is so (i.e. that it suffices to compute g_J^l in any one magnetic sublevel J_z), find the relevant g_J^l factors for the two multiplets $J = l \pm 1/2$.

(c) The above calculations are particular examples of a general result most easily proved using the Wigner-Eckart theorem. For a general vector operator $\vec{A}$ (one whose commutation relations with $\vec{J}$ are of the form $[J_i, A_j] = i\epsilon_{ijk}A_k$) the theorem says that

$$\langle J, M'|\vec{A}|J, M\rangle = \frac{\langle J|\vec{J}\cdot\vec{A}|J\rangle}{J(J+1)}\langle J, M'|\vec{J}|J, M\rangle$$

Use this theorem to rederive the g_J^l factor for $J = l + 1/2$ multiplet which you computed above.

Section B. Statistical Mechanics and Thermodynamics

1. An elastic string is found to have the following properties:

   - To stretch it to a total length x requires a force $f = \mu x - \alpha T + \beta T x$. Assume that α, β, μ are constants.
   - Its heat capacity at constant length x is proportional to temperature: $C_x = A(x)T$.

   We can use thermodynamic identities to derive from these facts a variety of other thermal properties. More specifically:

   (a) Calculate $\frac{\partial S}{\partial x}|_T$.

   (b) Show that A has to be independent of x.

   (c) Calculate $\frac{\partial S}{\partial T}|_x$ and give the general expression for entropy $S(x,T)$ assuming $S(0,0) = B$, where B is a constant.

   (d) Compute the heat capacity at zero tension $C_F = T\frac{\partial S}{\partial T}|_{f=0}$.

2. Let us model a white dwarf star as a degenerate Fermi gas of electrons, supported against gravitational collapse by the electron degeneracy pressure. For simplicity, we will assume that the star is a sphere of radius R and *uniform* mass density containing N electrons, N protons, and N neutrons for an approximate total mass of $M = 2Nm_p$.

   (a) First, assume that the electrons are not relativistic. Find their Fermi energy and show that at absolute zero, their total kinetic energy is

   $$U_k = \frac{3N(\hbar\pi)^2}{10m_e}\left(\frac{3N}{\pi V}\right)^{\frac{2}{3}}$$

   where V is the volume of the star. (Note that the total kinetic energy of the nucleons is much smaller than that of the electrons.)

   (b) The gravitational binding energy of a uniform-density sphere is

   $$U_{grav} = -\frac{3GM^2}{5R}$$

   Find the equilibrium radius for the white dwarf. Eliminate N. How does this radius depend on the mass?

   (c) If instead the electrons are highly relativistic, so that their energy and momentum are related by $\epsilon = cp$, then find the Fermi energy and show that the total kinetic energy is now

   $$U_k = \frac{3N\hbar\pi c}{4}\left(\frac{3N}{\pi V}\right)^{\frac{1}{3}}$$

   (d) Under what conditions is a highly relativistic degenerate electron star unstable against collapse? This is called the Chandrasekhar limit. A star that violates the limit will collapse into a neutron star or black hole, depending on whether neutron degeneracy pressure can hold up the star.

   Constants you may need:

   $$G/(\hbar c) = 6.707 \times 10^{-39}\ (\mathrm{GeV}/c^2)^{-2}$$

   $$1\ \mathrm{GeV}/c^2 = 1.78 \times 10^{-27}\ \mathrm{kg}$$

   $$M_\odot = 1.99 \times 10^{30}\ \mathrm{kg}$$

3. Consider a single free particle of mass m confined to a volume V. Let $Z_1(m)$ denote the quantum partition function for this system (where the partition sum is taken over the discrete energy levels of a particle of mass m in a box of volume V).

   (a) Show that $Z_1(m) \to V/\lambda^3$ with $\lambda = h/\sqrt{2\pi mkT}$ in the classical (or small $\hbar$) limit. Use this result to calculate the classical energy and heat capacity at fixed volume of the single particle system.

   (b) Identify the temperature at which this approximation breaks down.

   (c) Now consider a system consisting of two identical, non-interacting particles in the same box. Because of the effects of identical particle statistics, the classical expectation for the two-particle partition function $Z_2(m) = Z_1(m)^2$ is not quite correct. Show that the exact free boson and free fermion two-particle partition sums can in fact be expressed in a simple way in terms of the one-particle functions $Z_1(m)$ and $Z_1(m/2)$.

   (d) Using the classical approximation $Z_1(m) = V/\lambda^3$ derived in the first part of this problem, calculate the correction to the energy E and the heat capacity C due to Bose or Fermi statistics.

Department of Physics, Princeton University

Graduate General Examination
Part III

Monday, May 5, 2003
9:00 am - 12:00 noon

This part of the General Examination poses THREE questions on Condensed Matter Physics and THREE on Elementary Particles and Nuclear Physics. Answer THREE questions, at least ONE from each section.

Work each problem in a separate examination booklet. Be sure to label each booklet with your name, the section name, and the problem number.

Section A. Condensed Matter Physics

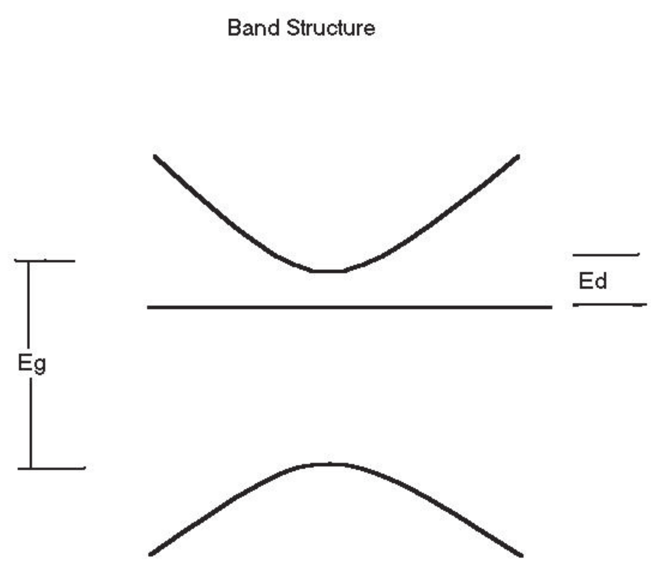
1. Consider a semiconductor with a band gap E_G and single conduction and valence bands with effective masses m_e and m_h respectively.

   (a) Calculate the equilibrium number densities of electrons n_e, and holes n_h in an intrinsic (i.e. undoped) semiconductor when the temperature is small compared to the band gap ($k_B T << E_G$).

   (b) What is the chemical potential μ of the electrons under these conditions? Take the precise center of the gap to be the zero of energy here.

   (c) Suppose instead that this semiconductor is now doped with a number density n_d of donors, each contributing one electron. Each donor can bind an electron with binding energy E_D (relative to the bottom of the conduction band). Calculate the number density n_e of electrons in the conduction band for temperatures low compared to E_G and E_D.

   The following figure illustrates the setup described in the problem:

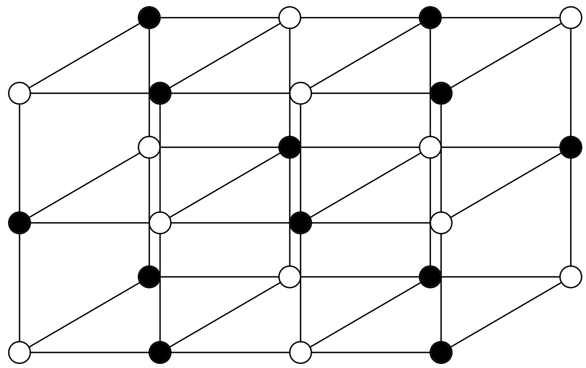


2. The bulk modulus of $NaC\ell$ is measured to be

$$V\frac{\partial P}{\partial V}|_{T=0} \;=\; 2.4\times 10^{10}\,\mathrm{N/m^2}$$

(where P is pressure V is volume).

The crystal structure of this salt is a simple cubic lattice, but with Na on half of the lattice sites and Cl on the other half, so that nearest neighbors are always different elements. The nearest-neighbor spacing is .28 nm.



● **Sodium Atoms**
○ **Chlorine Atoms**

Modeling the interaction responsible for the bulk modulus as nearest-neighbor springs, answer the following questions:

(a) What is the spring constant K?

(b) What is the resulting speed of *longitudinal* sound? This is the wave whose frequency goes to zero as the wavelength goes to infinity.

(c) What are the frequencies of the three optic modes? To remind you, the optic modes are vibrations where the frequency goes to a finite value and nearest neighbors oscillate out of phase in the limit in the limit as the wavelength goes to infinity.

Give answers in SI units and also give the quantum energies of the optic modes, $\hbar\omega_{optic}$, in ^oK. You will need to know the masses of the two atoms:

$$\mathrm{Na} = 23.\times(1.67\times 10^{-27})\mathrm{kg}, \qquad \mathrm{Cl} = 35.\times(1.67\times 10^{-27})\mathrm{kg}$$

3. Consider the Hubbard Model in two dimensions. The Hubbard Hamiltonian is

$$H = -t\Sigma_{i\sigma}\Sigma_j^{nn}\ a^+_{i+j\sigma}\ a_{i\sigma} + U\Sigma_i n_{i\uparrow}n_{i\downarrow}$$

Here $a^+_{i\sigma}$ creates an electron at site i and with spin σ and $n_{i\sigma} = a^+_{i\sigma}a_{i\sigma}$ is the usual number operator. A simple square lattice is assumed and the sum over j in the hopping term proportional to t is over the four nearest neighbor sites of i. We denote the lattice spacing by d.

(a) Calculate the energy spectrum of the single particle term in the Hamiltonian (the term proportional to t). Show that the bottom of the band lies at energy $-4t$ and calculate the density of states (per unit area) near the bottom of the band.

(b) Using the above density of states for a small number of electrons per atom calculate the Hartree-Fock energy for the paramagnetic state and the ferromagnetic state. Show that the ferromagnetic state has the lowest energy if U is above a critical value. What is that value?

(c) Show that the Fermi surface for the paramagnetic state with one electron per atom is a square with corners at $k = (\pm\pi/2d, 0)$ and $(0, \pm\pi/2d)$ when $U = 0$.

(d) Is the state just described (half-filled band) stable when U becomes non-zero? Explain your answer.

Section B. Elementary Particles and Nuclear Physics

1. The rates and branching ratios of elementary particle decays provided many important clues toward the establishment of the Standard Model. Here are a few questions about decays, to be answered in the light of our modern understanding of particle theory:

   (a) The η meson has mass 547 MeV, spin-parity 0^-, and strong isospin zero. It is classified as a "stable" particle despite the fact that it could, as far as mass is concerned, decay strongly into 2π or 3π. First, explain why $\eta \to 2\pi$ is forbidden for both strong and electromagnetic interactions. Second, explain why $\eta \to 3\pi$ is forbidden as a strong interaction but allowed as an electromagnetic interaction.

   (b) The muon has a mass of 106 MeV and a lifetime of 2.2 μs for its unique decay mode $\mu^- \to e^-\bar{\nu}_e\nu_\mu$. Its Standard Model third generation partner, the tau, has a mass of 1.78 GeV and the decay $\tau^- \to e^-\bar{\nu}_e\nu_\tau$ is observed to occur with a branching ratio of about 20%. Argue that the rate for this kind of three-body decay should scale as the fifth power of the mass of the decaying lepton. Use this result to evaluate the lifetime of the tau lepton.

   (c) The D^0 meson has quark content $(c\bar{u})$ and decays weakly. The relative branching ratios to apparently quite similar channels can be very different due to the intervention of Cabbibo factors in the weak interactions. Estimate the relative rates for the following two-body decay modes of the D^0:

   $$D^0 \to K^-\pi^+,\ \pi^-\pi^+,\ K^+\pi^-$$

   (Note that the quark content of the K^- is $(\bar{u}s)$).

   (d) A hadron is observed to decay to $\Lambda_c^+\ \pi^0$ with a decay width of about 2 MeV (indicating that it is a strong decay). Deduce the quark content of this hadron. Does it possess any isospin partners, and if so what are their quark contents and electric charges? The Λ_c^+ has quark content (udc) and is an isosinglet.

2. The particle physics community is lobbying for the construction of a new accelerator capable of producing e^+e^- collisions at center-of-mass energies of 500 GeV-1TeV. Experiments performed with this new collider would shed much light on the mechanism of electroweak symmetry breaking.

   (a) In the Standard Model, the Higgs boson couples to fermions through the interaction

   $$\mathcal{L} = \sum_i (\sqrt{2}G_F)^{1/2}\, m_i H \bar{\psi}_i \psi_i \,,$$

   where m_i is the mass of the i-th fermion and $G_F = 10^{-5} m_p^{-2}$ is the Fermi coupling (a convenient way of stating the strength of low-energy weak interactions). Calculate the decay width $\Gamma(H \to f\bar{f})$ where $f\bar{f}$ is a fermion-antifermion pair with $m_f \ll m_H$.

   (b) Precision electro-weak data and theoretical considerations (supersymmetry) favor a light Higgs boson. In fact, experimental hints from LEP briefly suggested a Higgs at $m_H = 115\,\text{GeV}$. At that mass, what would be the partial width $\Gamma_{e\bar{e}}$ for Higgs decay into an electron-positron pair via the above interaction? What would be the partial width $\Gamma_{b\bar{b}}$ for decay into a pair of b quarks? The relevant masses are $m_e = .5\,\text{MeV}$ and $m_b = 4.5\,\text{GeV}$.

   (c) A clear signal of the existence of the Higgs would be the resonance peak in the cross-section for

   $$e^+e^- \to H \to f\bar{f}$$

   for center-of-mass energy in the neighborhood of the Higgs mass. The Breit-Wigner formula for the resonance cross-section reduces in this case (after accounting for the spins of the colliding particles and of the resonance) to

   $$\sigma = \frac{4\pi}{m_H^2} \frac{\Gamma_{e\bar{e}}\Gamma_{f\bar{f}}}{\Gamma^2} \,.$$

   For a 115 GeV Higgs, the dominant channel is bottom-antibottom. What is the cross-section for this reaction at resonance? If a collider can achieve a luminosity of $5 \cdot 10^{33}\,\text{cm}^{-2}\text{sec}^{-1}$, how many events of this type would be produced per year, with the collider running at resonance.

3. In the Standard Model, the charged current interactions of the quarks take the form

$$\mathcal{L}^{CC} = -\frac{g}{\sqrt{2}} (\bar{u}_L\ \bar{c}_L\ \bar{t}_L)\, \gamma^\mu\, V_{CKM} \begin{pmatrix} d_L \\ s_L \\ b_L \end{pmatrix} W^\dagger_\mu + \text{h.c.}\,, \tag{1}$$

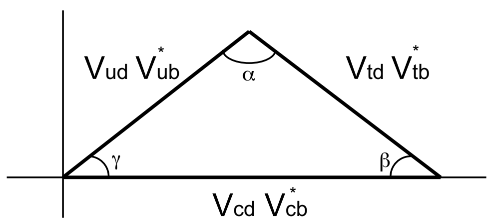
where the W_μ field corresponds to the W-boson and V_{CKM} (the Cabibbo-Kobayashi-Maskawa matrix) is a **unitary** matrix giving the amplitudes for weak transitions between the different flavors of quark. A more explicit representation of V_{CKM} is

$$V_{CKM} = \begin{pmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix}, \tag{2}$$

where the V_{ab} are complex numbers subject, by virtue of the unitarity of V_{CKM}, to constraints such as

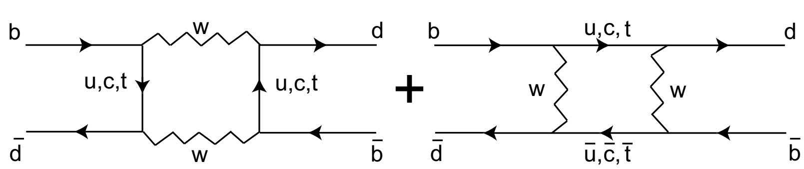
$$V_{cd}V^\star_{cb} + V_{td}V^\star_{tb} + V_{ud}V^\star_{ub} = 0$$

This constraint can be graphically summarized in 'unitarity triangles' such as



The phases of the V_{ab} are not individually meaningful: they can be changed by redefining the phases of the quark fields (viz. $u \to \exp(i\alpha)u$). If there is no way to choose the phases of the quarks such that all the V_{ab} are real, then CP (and T) are broken with all the consequences that follow. We will now explore some methods for determining the phases of elements of V_{CKM}.

Consider the neutral mesons $B^0 = \bar{b}d$ and $\bar{B}^0 = b\bar{d}$. $|B^0\rangle$ and $|\bar{B}^0\rangle$ are not mass eigenstates because they mix through the diagrams



To a very good approximation (this has to do with the GIM mechanism), these diagrams are dominated by those in which the internal lines are all top quarks. **Show**

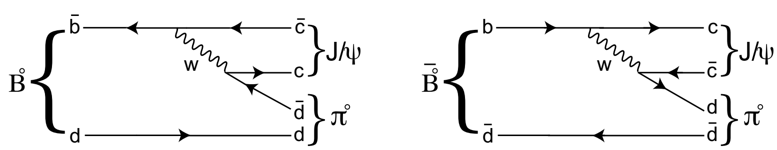
that both diagrams are proportional to $(V_{td}V_{tb}^\star)^2$. Assuming that all other contributions to the amplitude are real, the overall phase is determined by this CKM factor.

Neither B^0 nor $\bar{B}^0$ are mass eigenstates. To find the mass eigenstates, we must diagonalize a 2x2 Hamiltonian whose off-diagonal elements are given by the mixing amplitude. **Show that the energy eigenstates system are of the form**

$$|B_\pm\rangle = \frac{1}{\sqrt{2}}\left(e^{i\phi_M}|B^0\rangle \pm e^{-i\phi_M}|\bar{B}^0\rangle\right)$$

and show that ϕ_M **is related to the phase of the** $V_{td}V_{tb}^\star$ **leg of the unitarity triangle displayed above**.

We observe mixing by watching decays to a channel open to both B^0 and $\bar{B}^0$, such as $B^0(\bar{B}^0) \to J/\psi\,\pi^0$ which we assume to proceed via the following quark diagrams (we are simplifying reality here):



Identify the CKM factors associated with these diagrams and show that they correspond to another leg of the unitarity triangle shown above. In what follows we will assume that the complex phase ϕ_W of the decay amplitude comes entirely from the CKM factor.

Now suppose that a B^0 (or $\bar{B}^0$) is created at time $t = 0$. It will evolve into a linear combination of B^0 and $\bar{B}^0$ and the rate for decay to the $f = J/\psi\,\pi^0$ final state at later times t will have time-dependent interference effects. **Construct the time-dependent asymmetry**

$$a(t) = \frac{\Gamma(B^0 \to f) - \Gamma(\bar{B}^0 \to f)}{\Gamma(B^0 \to f) + \Gamma(\bar{B}^0 \to f)}$$

and show that by measuring it, one can determine a combination of the phases ϕ_M **and** ϕ_W.

Show that this combined phase is physically meaningful, i.e. is independent of redefinitions of the phases of the quark field. To what angle of the unitarity triangle does it correspond?

Department of Physics, Princeton University

Graduate General Examination
Part IV

Tuesday, May 6, 2003
9:00 am - 12:00 noon

This part of the General Examination poses SIX questions, TWO on Relativity and FOUR on General and Atomic Physics. You must do ONE relativity problem and TWO General and Atomic questions.

Work each problem in a separate examination booklet. Be sure to label each booklet with your name, the section name, and the problem number.

Section A. Relativity

1. Imagine a gigantic black hole of Schwarzschild radius $R_s = 3 \times 10^5$ km, described for $r > R_s$ by the metric

$$ds^2 = -\left(1 - \frac{R_s}{r}\right) dt^2 + \frac{dr^2}{1 - R_s/r} + r^2(d\theta^2 + \sin^2\theta d\phi^2) \ .$$

An unmanned craft starts from rest far away from the black hole and is in free fall. At $r = R_s + 10$ km, the craft passes a spaceship which keeps station at that radius using its rockets. At that instant, the clocks on the craft and the spaceship are set to zero and a device on the craft is activated which emits a pulse of frequency ν (as measured by the craft's clock) every 10^{-5} seconds. Thus, pulses are emitted at 10^{-5} s, 2×10^{-5} s, etc., according to the unmanned craft's clock.

   (a) How many pulses will the crew of the spaceship observe?

   (b) At what reading of the spaceship's clock will the last pulse be received?

   (c) What will the crew observe the frequency of the last pulse to be?

You are allowed to make reasonable approximations in answering all of the above.

2. In a universe with zero space curvature, mean mass density ρ, and cosmological constant Λ, the evolution of the scale factor $a(t)$ is described by the Friedmann equation:

$$H^2 \equiv \left(\frac{1}{a}\frac{da}{dt}\right)^2 = \frac{8\pi G\rho}{3} + \frac{\Lambda}{3}$$

where $H(t)$ is the Hubble expansion rate and G is Newton's constant of gravitation. This equation simply expresses conservation of energy (kinetic on the left and negative potential on the right). Assume $\Lambda \geq 0$.

(a) Define the critical density $\rho_c = 3H^2/(8\pi G)$ and the fractional density $\Omega \equiv \rho/\rho_c$. Note that Ω, ρ, and ρ_c all evolve with time. Rewrite the Friedmann equation in terms of a, $\dot{a} \equiv da/dt$, and the constants Λ, H_0, Ω_0, and a_0 (where the "0" subscript indicates present values).

(b) Start with the two opposite limiting cases $\Lambda = 0$ and $\rho = 0$. For each case, find the scale factor $a(t)$ and the current value q_0 of the deceleration parameter $q \equiv -a\ddot{a}/\dot{a}^2$. In each case, make a reasonable choice for the integration constant and explain it.

(c) At what redshift does the universe change from being matter dominated to vacuum energy dominated? Answer in terms of Λ, H_0, Ω_0.

(d) Show that the following solves the full Friedmann equation you found in part 2a:

$$a(t) = K \sinh^{2/3}\left(\frac{\sqrt{3\Lambda}}{2}t\right).$$

Find an expression for K, again in terms of the same constants.

(e) Find an expression for $H(t)$ in terms of $\Omega(t)$. You can use it to eliminate Λ in the following.

(f) Suppose $(H_0)^{-1} = 13.8 \times 10^9$ years, usually called the "Hubble Time". What age of the universe is implied if matter makes up the entire critical density, that is if $\Omega_0 = 1$? What age of the universe is implied if instead $\Omega_0 = 0.27$, as found by the WMAP satellite and other observations?

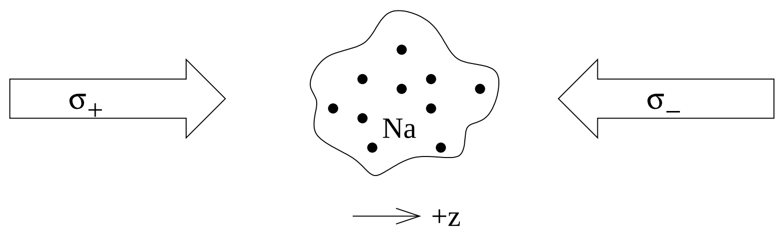
Section B. General and Atomic Physics

1. A magneto-optical trap (MOT) is commonly used for a variety of experiments with alkali metal atoms. The MOT can both cool the atoms and trap them in a small region. The trap has a magnetic field gradient and several intersecting circularly polarized laser beams.

   Many features of the MOT can be found by considering only one dimension at a time, as you will do here. The 1-D trap has two right-circularly-polarized (RCP) laser beams of the same frequency coming from left and right (labels σ_+ and σ_- give the polarization direction relative to the $+z$ axis), as well as a linearly varying magnetic field oriented along the z-axis:

   $$\mathbf{B}(z) = \beta_0 \mathbf{z}$$

   Let the magnetic field gradient $\beta_0 > 0$. (A real trap has a field gradient in all 3 dimensions and 6 lasers approaching along all Cartesian axes.)



   Consider a trap designed for sodium ($Z = 11, A \approx 23$) atoms. The unpaired electron in the sodium ground state is ${}^2S_{1/2}$, while the lowest two excited states are ${}^2P_{1/2}$ and ${}^2P_{3/2}$. The transitions from the ground state to the ${}^2P_{1/2}$ and ${}^2P_{3/2}$ states are called the D_1 and D_2 transitions, respectively. The transitions have vacuum wavelengths of $\lambda_1 = 589.757$ nm and $\lambda_2 = 589.158$ nm. The excited states have lifetimes $\tau_1 = 16.25$ ns and $\tau_2 = 16.30$ ns. The Bohr magneton is $\mu_B = e\hbar/2m_e = 5.79 \times 10^{-9}$ eV/gauss. In this problem, ignore the nuclear spin of sodium.

   (a) First consider Doppler cooling of sodium atoms. For this section, let the magnetic field be turned off. How should the lasers be tuned relative to the sodium atom transition frequencies to cool the atoms (i.e. reduce the spread in their velocities) along the z-axis? Give a rough estimate of the maximum velocity the atoms can have to be cooled in this arrangement. Also estimate the number of photons that must be absorbed to cool the fastest atoms. Will the cooling process be more efficient if the lasers are tuned near the D_1 or the D_2 transition?

   Because of photon recoil the cooled atoms still have non-zero temperature and velocity, so we need some other method to confine them for long times. For this purpose, we restore the magnetic field and take advantage of the Zeeman splitting of the sodium orbitals.

(b) Sketch the energy levels of the sodium atom in an external magnetic field, indicating all $|J, m_j\rangle$ components of the ground state and first two excited states. Give the Zeeman splitting (MHz/gauss) of the magnetic sublevels within each state for small fields. The g-factors for the states are given by

$$g(j, \ell, s) = 1 + \frac{j(j+1) - \ell(\ell+1) + s(s+1)}{2j(j+1)}$$

(c) Because of the magnetic field gradient, the Zeeman splitting is position-dependent. Sketch the transition frequencies as a function of z in the varying magnetic field. What is the direction of the net force experienced by an atom at rest some distance away from $z = 0$?

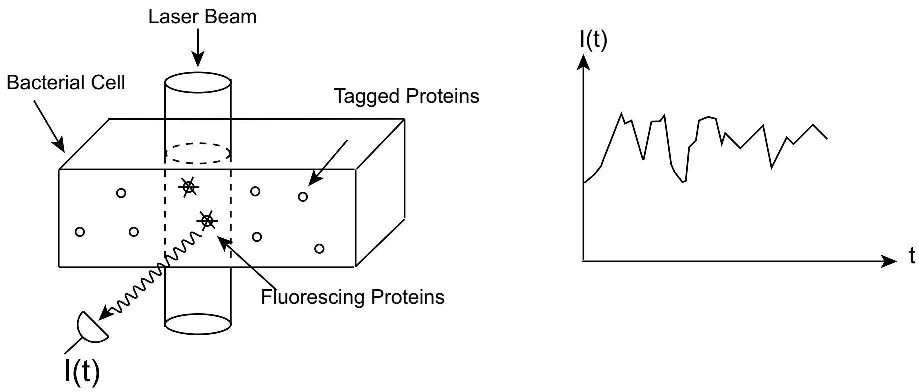
(d) Estimate the magnetic field gradient required to keep atoms in a trap of width $r \sim 1$ cm.

2. Write a brief description of the following sources of electromagnetic radiation. Your description should include : a) an estimate of the mean wavelength of the radiation; b) an estimate of the power radiated; c) as specific a description as you can manage of the physical mechanism by which the radiation is generated. Include a brief explanation of the reasoning you used to make your estimates.

   (a) A 100 Ci source of ^{60}Co gamma rays (1 Ci $= 3.7 \times 10^{10}$ disintegrations per second).
   (b) A household microwave oven.
   (c) The screen of a color television set.
   (d) A laser pointer for a lecture.
   (e) The sun.
   (f) The earth as a radiator to space.
   (g) A dental X-ray machine.

   Do your best to 'guesstimate' any constants that may be needed to answer these questions. You may not be able to be able to answer all of them at the same level of detail, but do as much as you can.

3. We can look into a living cell with a microscope and see very small numbers of molecules if these molecules can be made to fluoresce. Some of the cell's own proteins can be made fluorescent by means of chemical or genetic tricks and it would be interesting to count them, cell by cell, by measuring their fluorescence. While the fluorescence signal is proportional to the number of molecules under illumination, it can be difficult to measure the proportionality constant in an independent experiment. It is therefore not so easy to determine absolute concentrations of molecules in a cell by this method. Cluzel et al. (in experiments carried out at Princeton) circumvented this problem by watching small numbers of molecules diffusing randomly in and out of an illuminated volume inside an individual cell and using the *variance* in the fluorescence intensity, along with its mean value, to make an absolute measurement of the concentration of the molecules.



(a) Explain (qualitatively) how this measurement might work. What do you gain by using both the variance and the mean of this signal?

(b) Explain (qualitatively) how the fluctuating fluorescence signal could be analyzed further to give an estimate of the diffusion constant of the fluorescent protein molecules.

(c) Now let's convert the above intuition into a quantitative framework for analysis of the data. Consider the concentration $c(\vec{x}, t)$ of fluorescent molecules at different points in space and time. It fluctuates and the deviation δc of the concentration from its average value $\bar{c}$ is uncorrelated between different points in space (but the same instant of time). Show that the analytic statement

$$\langle \delta c(\vec{x},t)\delta c(\vec{x}',t)\rangle = \bar{c}\delta(\vec{x}-\vec{x}')$$

of this fact is equivalent to the 'intuitive' remark that the variance of the number of molecules in a volume is equal to the mean number.

(d) If the system starts with some fluctuation in the concentration $c(\vec{x},0) = \bar{c} + \delta c(\vec{x},0)$, this profile will relax according to the diffusion equation. Since the diffusion equation is linear, this means that the profile of fluctuations at time t, $\delta c(\vec{x},t)$, can be written as a linear operator acting on the initial condition $\delta c(\vec{x},0)$. Show that this linear relationship can be written as

$$\delta c(\vec{x},t) = \int d^3y \left(\frac{1}{\sqrt{4\pi Dt}}\right)^3 \exp(-|\vec{x}-\vec{y}|^2/4Dt)\ \delta c(\vec{y},0)$$

where D is the diffusion constant.

(e) When we bring light to a focus under the microscope, we effectively weight the points around the focus with a Gaussian function, so that the light intensity collected from the fluorescent molecules will be proportional to

$$s(t) = \int d^3x c(\vec{x},t)\exp(-|\vec{x}|^2/\lambda^2)$$

where λ is the size of the focal region (roughly the size of the wavelength of light). Using the result of (d), show that the temporal correlation function of this signal is given by

$$\langle \delta s(t)\delta s(0)\rangle \propto (|t|+\tau)^{-3/2}$$

and relate the correlation time τ to the diffusion constant D and the size of the focal region λ. (Hint: in doing the multi-dimensional Gaussian convolution integrals that show up in the last step of this computation, it is a good idea to do them Cartesian coordinate by Cartesian coordinate!) This gives a precise method for extracting the diffusion constant from the fluctuating fluorescence signal.

4. The expression of genes in the cell is regulated by the binding of special proteins to particular sites (typically 10-20 base pairs long) on the DNA. These proteins are present in modest numbers in the cell, but nonetheless control the state of the cell. In first approximation, they can be either in solution at chemical potential μ, or bound to a single DNA site with energy $-E$. We will examine the kinetics of their binding and unbinding to see how fast the cell can respond to external influences.

   (a) Use the grand canonical partition sum over the two possible states for the site (empty vs. once occupied) to find the equilbrium probability of the site being occupied. Express your answer as a function of the chemical potential μ, the binding energy $-E$ and $\beta = 1/k_B T$.

   (b) This result is not very useful unless we know how μ depends on the concentration c of the protein in the solution. So, let's assume the dilute solution result $\mu = k_B T \ln(c/c_0)$ (where c_0 is some reference concentration). Use this to rewrite the occupation probability of the site as a function of c (you should find that it rises from 0 to 1 as c increases). Obtain an expression for $c_{1/2}$, the concentration where the site is occupied with 50% probability.

   (c) Protein molecules bind to the site at a rate $k_+ c$ which of course increases with the concentration of molecules in solution. Bound molecules fall off the DNA at a rate k_-. In an ideal (dilute) solution, both k_+ and k_- are independent of concentration. Show that detailed balance in equilibrium implies that $k_+ c_{1/2} = k_-$.

   (d) Let's look into the physics of this more closely. The binding rate k_+ is limited by diffusion. Give a simple dimensional analysis estimate of the "diffusion limited" rate k_+ in terms of the diffusion constant D of the protein and the geometrical size a of the site it must find. What is the numerical value of k_+ under the reasonable assumptions that $D \sim 3\mu m^2/sec$ (the right number for a sphere of typical protein size diffusing in water) and $a \sim 1nm$ ($10^{-7}cm$) (the rough size of a length of DNA containing 10 consecutive base pairs)?

   (e) In a bacterium of volume 1 $(\mu m)^3$, there might be 100 protein molecules of the relevant type. Assume that a total of 50 molecules in the cell gives a concentration large enough that the site is occupied with probability one half: then with 100 molecules present, one will be bound and 99 will be free. If all of the 99 free molecules are suddenly eaten up by some process in the cell, how fast can the one bound molecule fall off the site and hence change the expression rate of the gene being controlled? If your answer is biologically sensible, it will be neither too short (microseconds) nor too long (years)!

Stanford University 2005 Qualifying Examination

General Physics Problem I

Ground-based optical astronomers are familiar with a phenomenon called Atmospheric Dispersion, wherein stars observed away from the zenith (i.e. the point directly overhead) appear to be distorted as a function of color, due to refraction in the atmosphere. The purpose of this problem is for you to estimate the magnitude of this effect from first principles. The logic and consistency of your physical arguments is more important than the accuracy of your numerical answers.

(a) Since physical conditions vary significantly with altitude in the atmosphere, you might expect this effect to be sensitive to the details of the atmospheric models. Show, however, that to a good approximation, the difference in the apparent position of a star at a given wavelength, λ, and its true position, depends only on the zenith angle (the angular distance away from the zenith) and the value of the index of refraction of the atmosphere for that wavelength at the altitude of the observatory. Write an explicit expression for the angular spread $\Delta\theta(\lambda_1, \lambda_2)$ due to atmospheric dispersion for a wavelength range $\lambda_1 - \lambda_2$, as a function of the index of refraction evaluated at those two wavelengths and the zenith angle, θ.

(b) Treat the atmosphere as a dilute gas of molecules which can be approximated as classical harmonic oscillators with resonant frequency, ω_0. Write (or derive) an expression for the index of refraction as a function of frequency, ω, in terms of ω_0, the molecule number density, N, and any other physical constants you think are required.

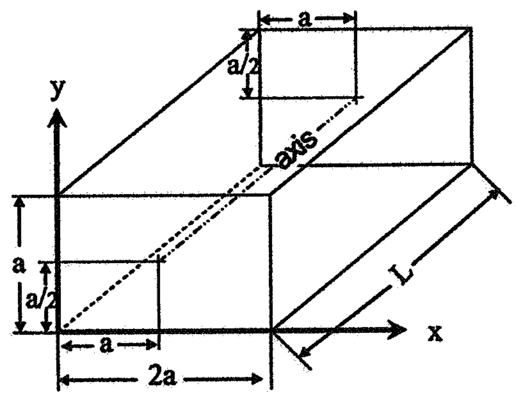
(c) Giving simple physical arguments, estimate appropriate values for ω_0 for air molecules, and for the number density, N, at standard temperature and pressure. Use these to estimate the index of refraction of air at the extremes of the optical band: $\lambda_{min} = 400$ nm, $\lambda_{max} = 900$ nm.

(d) Typical modern observatories are located at altitudes of ~ 3000 m above sea level. How do you expect the index of refraction values at this altitude to compare to your answer from part (c)? Give a quantitative estimate.

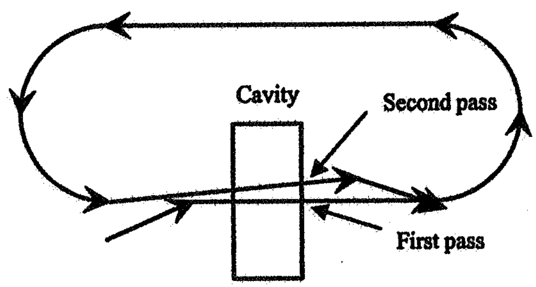
(e) Use your quantitative answers above to determine the approximate angular size of an unfiltered star image due to atmospheric dispersion alone, when it is observed at a zenith angle of 60°.

2005

Electricity and Magnetism



1) A rectangular microwave cavity is excited in the lowest frequency mode that has no magnetic field in the longitudinal (z) direction and a zero of the longitudinal electric field on the axis. The cavity has dimensions a x 2a x L. The axis is defined by the line $x = a$, $y = a/2$, running through the center of the cavity.
   a) If $a = (4.5)^{1/2}$ cm $= 3/2^{1/2}$ cm, what is the frequency?
   b) Give an expression for each component of the electric and magnetic fields.

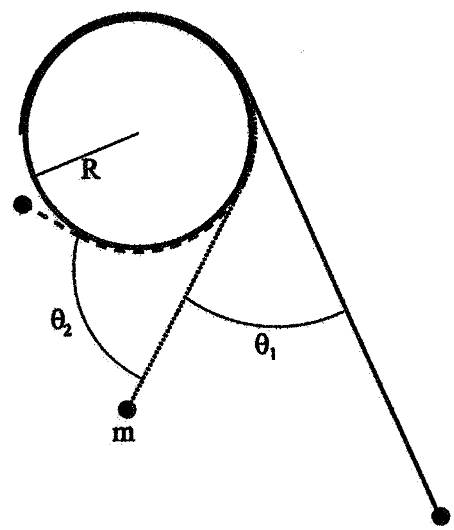


2) A continuous beam of relativistic electrons ($E = \gamma m_0 c^2$) passes through the cavity and is bent back around by a system of magnets so that it passes though the cavity again, a time T later. As suggested in the figure, the first pass is on the cavity axis, while the second pass is off the axis solely due to the influence of the cavity fields on the first pass. The only fields in the cavity are those due to the mode above. The beam current is I and the cavity quality factor is Q ($= \omega U/P_{cavity}$, where U is the energy stored and P_{cavity} is the power dissipated in the cavity). Assume that as far as the electron beam is concerned, γ is a constant, and the cavity is short enough that its influence on the beam can be treated as an impulse.

   Show that if the current is greater than some value, $I_{threshold}$, the fields can extract energy from the beam and grow exponentially. Below this value the fields will decay to zero.

Classical Mechanics, 2004-05

A pendulum is constructed by attaching a mass m to an extentionless string of length l. The fixed end of the string is connected to a fixed vertical disk of radius $R(R < l/\pi)$, as shown in the figure below. The motion of the pendulum occurs in the plane of the figure and the string connection to the disk is at a place such that in the oscillations of interest here the string is always tangent to the disk. Find the equation of motion for the pendulum and calculate the period of the small oscillations. **In the small oscillations regime** find the line about which the angular motion extends equally in either direction (i.e. $\theta_1 = \theta_2$).



QM I

QUALIFIER EXAM 2004–5 QUANTUM MECHANICS QUESTION

Consider the Hilbert Space describing the spin degrees of freedom of three distinguishable spin 1/2 particles. Denote Pauli matrices acting on particle a as σ_i^a where $a = 1, 2, 3$ and $i = x, y, z$. Consider the hermitian operators

$$\begin{aligned} A &= \sigma_x^1 \sigma_y^2 \sigma_y^3 \\ B &= \sigma_y^1 \sigma_x^2 \sigma_y^3 \\ C &= \sigma_y^1 \sigma_y^2 \sigma_x^3 \\ D &= \sigma_x^1 \sigma_x^2 \sigma_x^3 \end{aligned}$$

- These operators mutually commute so they can be simultaneously diagonalized. What are the eigenvalues of these operators? Show they obey the operator equation

$$ABCD = -I \tag{1}$$

In the theory of quantum measurement one often contrasts quantum mechanics with a classical "hidden variable" theory. Roughly speaking, in such a theory measurement assigns an ordinary numerical values to any possible observable. If the observable is O then denote its assigned value $v(O)$. Basic consistency requires that $v(O)$ must be equal to one of the eigenvalues of the operator O, the results of quantum mechanical measurement. Any operator equation obeyed by mutually commuting observables will be obeyed by their eigenvalues and hence for consistency must also be obeyed by the values assigned by v. For instance the operator equation (1) implies the value equation $v(A)v(B)v(C)v(D) = -1$. You will now show that no such value function v can exist.

Consider the ten operators $A, B, C, D, \sigma_x^1, \sigma_y^1, \sigma_x^2, \sigma_y^2, \sigma_x^3, \sigma_y^3$ and their values $v(A), v(B), v(C), v(D), v(\sigma_x^1), v(\sigma_y^1), v(\sigma_x^2), v(\sigma_y^2), v(\sigma_x^3), v(\sigma_y^3)$.

- Show that the following operator equations hold.

QM I

$$
\begin{aligned}
I &= A\sigma_x^1\sigma_y^2\sigma_y^3 \\
I &= B\sigma_y^1\sigma_x^2\sigma_y^3 \\
I &= C\sigma_y^1\sigma_y^2\sigma_x^3 \\
I &= D\sigma_x^1\sigma_x^2\sigma_x^3 \\
-I &= ABCD
\end{aligned}
$$

Each of the above equations involves a mutually commuting set of operators so they each give rise to an equation on values. Now use these value equations to show that no value function v can exist.

This gives a strikingly simple argument that local realism is inconsistent with quantum mechanics.

2005 QM II

Consider a particle of mass m in a double well potential of the form

$$V(x) = V_0 L^{-4}[x^2 - L^2]^2.$$

a) The only obvious dimensionless quantity that can be made using the parameters of this problem is

$$A = L^2 V_0 m / h^2.$$

Give a physical interpretation to this parameter.

b) In the limit that $A >> 1$, make an estimate of the ground-state energy, E_0, and the difference in energy, ΔE, between the ground-state and the first excited state. Express all energies in units of V_0. You do not have to get numerical coefficients correctly, but you do need to exhibit correctly the dominant dependence of the answer on A in the limit of large A.

c) Draw a picture of the ground-state and first excited state wave-functions.

d) In the limit that $A << 1$, make an estimate of the ground-state energy and the difference in energy between the ground-state and the first excited state. Again, express all energies in units of V_0. You do not have to get numerical coefficients correctly, but you do need to exhibit correctly the dominant dependence of the answer on A in the limit of small; A.

e) Draw a picture of the ground-state wavefunction.

Clarification: What we are after in problems b and d is the leading A dependence. If, for instance, at large A, the ground energy were $E_0 = 6.417\, A^6 \exp[\, 43.91\, A\,]$ (which it isn't), then an answer worthy of full credit would be $E_0 \sim A^x \exp[\, \alpha A\,]$ with the notation that α and x are pure numbers of order 1. If the answer were $E_0 = 6.7\, A^4$, full credit would be awarded for $E_0 \sim A^4$.

Statistical mechanics qual exam January 2005

1 Part A

A uniform system has N particles in a volume V. The Helmholtz free energy F is given by $F = E - TS$, where E is the internal energy, T is the absolute temperature, and S is the entropy. Use the first and second laws of thermodynamics to obtain the following results:

$$S = -\left(\frac{\partial F}{\partial T}\right)_{VN}, \quad p = -\left(\frac{\partial F}{\partial V}\right)_{TN}, \quad \mu = \left(\frac{\partial F}{\partial N}\right)_{TV},$$

where μ is the chemical potential and p is the pressure.

2 Part B

An ideal gas consists of N indistinguishable ultrarelativistic classical particles (each has the dispersion relation $\epsilon_p = cp$ where $p = |\vec{p}| = \sqrt{p_x^2 + p_y^2 + p_z^2}$ and c is the speed of light in vacuum)

1. Calculate the partition function $Z(T, V, N)$ for this gas.
2. Use Stirling's approximation $\ln N! \approx N \ln N - N$ to derive the following expression for the Helmholtz free energy

$$F = -k_B T N \left[\ln\left(\frac{V}{N\lambda_R^3}\right) + 1\right],$$

   where $\lambda_R = \pi^{2/3}\hbar c/(k_B T)$ is the relativistic thermal wavelength. How does the classical limit constrain λ_R relative to the interparticle spacing $\ell \equiv (V/N)^{1/3}$?
3. Use part A to find the equation of state and the chemical potential.
4. Similarly, find the entropy and the heat capacity C_V at constant volume. How and why does C_V differ from that for an ideal nonrelativistic classical gas?
5. Find the heat capacity C_p at constant pressure. Find $C_p - C_V$ and compare with the nonrelativistic result.

2005

Special Relativity

Proton-antiproton annihilation

Consider an antiproton of mass m moving with initial velocity v and experiencing a collision with a proton which is at rest in the laboratory reference frame (a target). This collision leads to their annihilation, with emission of two photons.

1) Explain why annihilation produces two photons rather than one.

2) What is the velocity u of each of the particles in the center of mass system?

3) In terms of v and m, what is the energy w_{cm} of each of the two photons created in the annihilation process in the center of mass system?

4) Assuming the photons are emitted at an angle θ to the direction of motion in the center of mass frame, find the energies of the photons in the laboratory frame, in terms of v, m and θ.

5) For $\theta = 0$ (photon emission in the direction of motion of the anti-proton) find the laboratory frame energy of the least energetic photon in the limit $v \to 0$, and in the limit $v \to c$.

2005

General Physics II

There was a lot of interest this past fall in SpaceShipOne's successful bid for the X-prize. At the time of writing, not all the specs of the craft were public. It was launched from the aircraft 'White Knight' at an altitude of 15km. We know that it has a hybrid fuel (solid rubber propellant, liquid N_2O 'laughing gas' oxidizer) rocket motor, which, unlike a regular solid fuel motor can be shut down in flight, before burnout. We also know that about 2/3 of the total weight of the spacecraft was propellant and that a full burn-through takes 90s. On the first flight to prize altitude (100km), the ship had a ~75s equivalent burn.

a) Assume that the burn rate is constant, and that the craft uses its initial release velocity to pull to vertical, so that it starts nearly at rest. Derive an expression for the maximum altitude as a function of the burn time $t_b \leq 90$s, the initial altitude h_0, and the exhaust speed v_{ex} of the propellant. Neglect wind resistance.

b) For the launch parameters and total fuel fraction above, give the value of v_{ex} that gets you to the required 100km if you burn for 75s. For the answer, you will need to solve a quadratic equation; I'd recommend evaluating the coefficients numerically as otherwise the algebra is a bit tedious.

c) How high would you go if you burned the full 90s?

d) Looking at your equation in a) it is easy to see that the reason for a high altitude plane launch is not h_0. It is, of course, to reduce wind resistance. Assume an isothermal room temperature nitrogen atmosphere. How much smaller is wind resistance at 15km than at seal level? (Spaceship one is a pretty stubby craft, so blunt object drag is good enough – note you might want to refer to, but will not actually need to use, the drag formula).

University of Chicago, Autumn 2005
Ph.D. Candidacy Examination

Autumn 2005

DEPARTMENT OF PHYSICS
Ph.D. CANDIDACY EXAMINATION

Day 1
Wednesday September 14, 2005
(Problems 1 - 6)

Work all six problems. Please write clearly and show all the steps of your work. Define any symbols that you introduce. Credit will be given only for significant progress toward a solution. Give numerical answers when asked to do so. Use clear diagrams wherever appropriate.

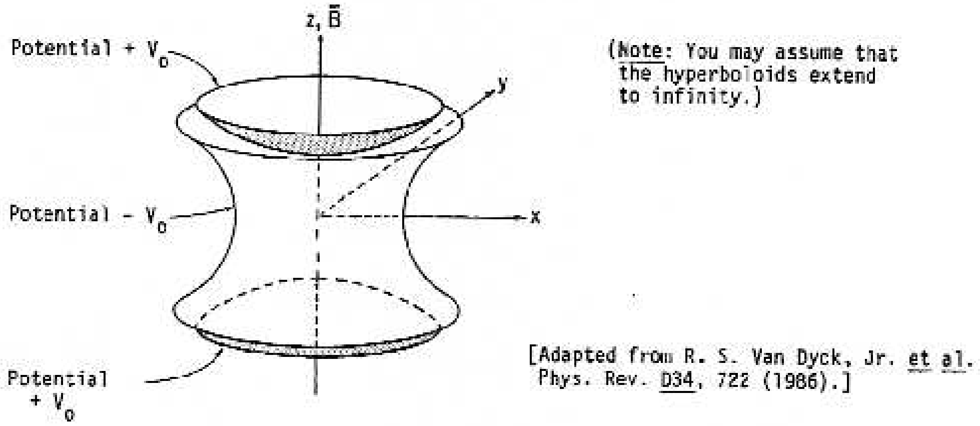
NO NAMES SHOULD APPEAR ON ANYTHING YOU SUBMIT; USE YOUR CODE NAME/NUMBER ONLY.

1. Relativistic dynamics.

(a) At a given instant in time the velocity of a particle of rest mass m is given by $\mathbf{v} = v\hat{i}$. Setting the speed of light to unity ($c = 1$) show that, in special relativity, the 3-force on this particle is not in the same direction as its 3-acceleration.

(b) A relativistic particle of mass m_1 and initial total energy E_1 strikes a stationary particle of mass m_2. The particles stick together and form a single particle of mass M that moves in the direction of the initial particle. Obtain an expression for the mass M in terms of m_1, m_2, and E_1.

(c) Determine what M from part (b) becomes in the nonrelativistic limit.



2. A *Penning trap* consists of a pair of electrodes at a potential $+V_0$ which form hyperboloids of revolution

$$z^2 - \beta^2 \rho^2 = z_0^2 \tag{1}$$

and an electrode at a potential $-V_0$ which forms a hyperboloid of revolution

$$\rho^2 - \gamma^2 z^2 = \rho_0^2 \quad . \tag{2}$$

Here $\rho \equiv (x^2 + y^2)^{1/2}$ is the radial coordinate and z is the axial coordinate, in a cylindrical coordinate system. A uniform magnetic field $\vec{B}$ is also applied along the z axis. Assume that the electrodes extend to infinity.

(a) Find the restrictions on the parameters β, γ, z_0, and ρ_0 in (1) and (2) in order that the potential between the electrodes be a function of ρ and z of the form: $\phi(\rho, z) = cz^2 - d\rho^2$, and give explicit expressions for c and d.

(b) Assuming the values of the parameters in part (a), find the angular velocity of an electron in an equilibrium circular orbit with $z = 0$ and $\rho =$ constant, and show that this orbit is stable for suitable V_0.

3. The most general state of two distinguishable spin-1/2 particles is

$$\Psi = a \left|\uparrow\right\rangle_1 \left|\uparrow\right\rangle_2 + b \left|\uparrow\right\rangle_1 \left|\downarrow\right\rangle_2 + c \left|\downarrow\right\rangle_1 \left|\uparrow\right\rangle_2 + d \left|\downarrow\right\rangle_1 \left|\downarrow\right\rangle_2$$

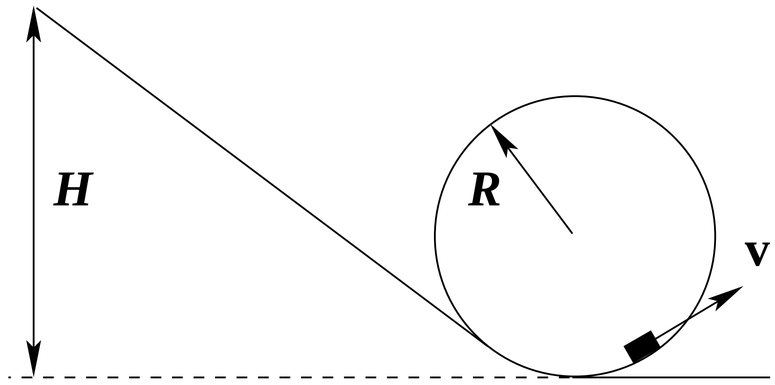
where $\left|\uparrow\right\rangle$ and $\left|\downarrow\right\rangle$ denote z-spin eigenstates and $|a|^2 + |b|^2 + |c|^2 + |d|^2 = 1$. Suppose the system is in the state

$$\Psi_0 = a_0 \left|\uparrow\right\rangle_1 \left|\uparrow\right\rangle_2 + b_0 \left|\uparrow\right\rangle_1 \left|\downarrow\right\rangle_2 + c_0 \left|\downarrow\right\rangle_1 \left|\uparrow\right\rangle_2 + d_0 \left|\downarrow\right\rangle_1 \left|\downarrow\right\rangle_2$$

If one were able to make any measurement one desired about the system, i.e., the spins of both particles about arbitrary directions, then it is well known that the only states that would give the same probabilities for all the possible outcomes as Ψ_0 would differ from Ψ_0 at most by an overall phase, i.e., $a = e^{i\alpha}a_0$, $b = e^{i\alpha}b_0$, $c = e^{i\alpha}c_0$, $d = e^{i\alpha}d_0$ for some real α. However, if less can be measured, then more states Ψ will give the same probability as Ψ_0 for the probabilities of what can be measured. Find the conditions on a, b,c, and d such that Ψ will give the same probabilities as Ψ_0 for the outcomes of measurement of

(a) The z-spin of both particles.

(b) The spin of particle 1 in an arbitrary direction.



4. An inclined plane is smoothly connected to a vertical loop of radius R. A small body, initially at rest, is released from an initial height H on the plane and then slides down and enters the loop.

(a) Assume that there is no friction. What is the minimum height H_{min} such that the body remains in contact with the loop at the loop's highest point?

(b) Now assume that there is friction between the loop and the sliding body, with the coefficient μ, while the inclined plane remains frictionless. Answer the same question as in part (a).

5. The Δ, with a mass of about 1232 MeV (we take $c = 1$), is the lowest-lying resonance formed by a pion π [$M(\pi) \simeq 138$ MeV] and a nucleon $N = (p \text{ or } n)$ [$M(N) \simeq 939$ MeV], where the masses are averages over charge states. The *isospins* I (isospin is a conserved internal SU(2) quantum number analogous to angular momentum) of these particles and the corresponding components I_3 are shown in the Table.

Particle	I	I_3	Particle	I	I_3
π^+	1	1			
π^0	1	0	Δ^{++}	3/2	3/2
π^-	1	-1	Δ^+	3/2	1/2
			Δ^0	3/2	$-1/2$
p	1/2	1/2	Δ^-	3/2	$-3/2$
n	1/2	$-1/2$			

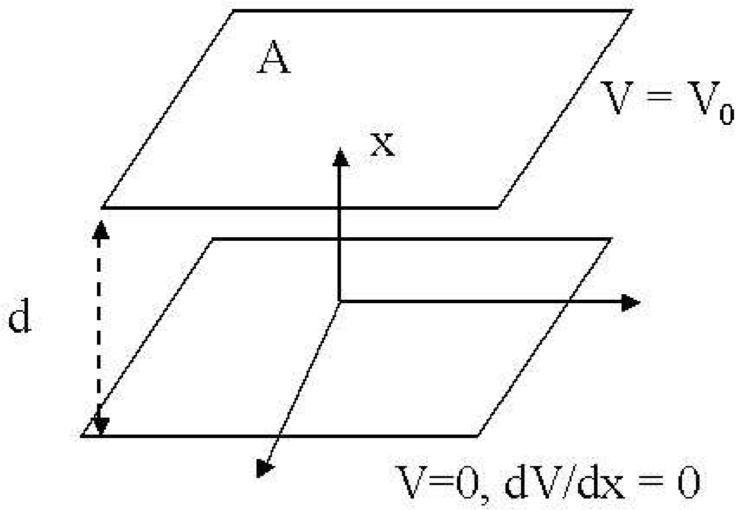
(a) Calculate the ratio of the cross sections for the processes

$$\pi^+ + p \to \quad \Delta^{++} \quad \to \pi^+ + p \ , \tag{3}$$

$$\pi^- + p \to \quad \Delta^0 \quad \to \pi^- + p \ , \tag{4}$$

$$\pi^- + p \to \quad \Delta^0 \quad \to \pi^0 + n \ . \tag{5}$$

(b) Generalize the results in part (a) to reactions of $\pi^\pm$ on neutrons. How would you study these processes?



6. Ohm's law is violated in a vacuum diode. The device consists of parallel plates of area A and separation d, with an electric potential V_0 across the plates. The plate area A is much larger than d^2. The cathode (the electrode at $V = 0$) should be considered to be an unlimited source of electron "gas", which fills the entire region between the two plates. These free electrons cancel the total potential near the cathode so that the boundary conditions are $V(x = 0) = 0$, $V(x = d) = V_0$, and $\mathrm{d}V/\mathrm{d}x(x = 0) = 0$.

If the electron current is **not** significantly affected by the collisions, derive the relation between the current I and V_0.

Autumn 2005

DEPARTMENT OF PHYSICS
Ph.D. CANDIDACY EXAMINATION

Day 2
Thursday September 15, 2005
(Problems 7 - 12)

Work all six problems. Please write clearly and show all the steps of your work. Define any symbols that you introduce. Credit will be given only for significant progress toward a solution. Give numerical answers when asked to do so. Use clear diagrams wherever appropriate.

NO NAMES SHOULD APPEAR ON ANYTHING YOU SUBMIT; USE YOUR CODE NAME/NUMBER ONLY.

7. A particle of mass m is subject to a central force

$$F = -\frac{k}{r^3} \ .$$

For which values of the conserved quantities is the motion of the particle bounded?

8. An electric dipole consisting of two charges $+q$ and $-q$ separated by a fixed distance d is placed at a distance L from a large, flat, grounded, conducting plane ($L \gg d$).

(a) Determine the orientation(s) for which the dipole experiences no net torque. Determine the stability of the zero-torque orientation(s).

(b) If the initial orientation of the dipole is random, estimate the mean size of the electric field at the conductor surface at the closest point to the dipole.

9. N spin-1/2 particles of mass m are placed in a two-dimensional harmonic potential

$$V(x,y) = \frac{m\omega^2}{2}(x^2 + y^2)\ .$$

You may assume that $N \gg 1$ and that the temperature is zero.

(a) What is the ground state energy of this system?

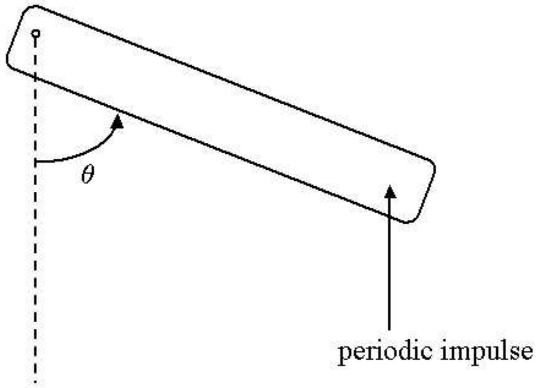
(b) What is the Fermi energy of this system?

10. Suppose that you are given a piece of a two-dimensional semiconductor, a current source, and a voltmeter.

(a) Using these components, how would you detect an external magnetic field?

(b) The density of electrons in the semiconductor is 1.00×10^{15} m^{-2} with an uncertainty of 0.5% coming from spatial inhomogeneity. The current source is a digital instrument with 8-bit resolution with the maximum output of 1 mA. The voltmeter, which has a 10-bit resolution, can read up to 1 V. Determine the sensitivity of your set-up for magnetic field detection for voltage and current. Also determine the absolute uncertainty in the magnetic field measurement.

(c) Will you be able to detect the earth's magnetic field?



11. Consider a bar of length ℓ fastened at one end to a frictionless pivot. It has a moment of inertia I about the pivoted end of the bar. The other end is subjected to a vertical periodic impulsive force, which imparts a change of angular momentum K/ℓ at each kick. The impulse is applied at times $t = 0$, τ, 2τ, ... (There is no gravity.)

(a) How are the angular momentum p_{n+1} and the angular position of the rotor θ_{n+1} after the $n+1$th kick related to the angular momentum p_n and position θ_n after the nth kick?

(b) Consider a limit of $K = 0$ in which the periodic kicks delivered do not change the total energy of the rotor even as the number of kicks goes to infinity. The only effect of the kick is to act as a "strobe" allowing us to take snapshots of the rotor position and momentum at fixed intervals in time. Is the typical motion periodic or aperiodic? Explain why.

(c) When K is very large, the rotor motion becomes apparently random. Suppose θ_n can be treated as effectively random, uniformly distributed, and uncorrelated at different times (i.e. at different n). Now consider a large number of different runs of the same rotor, corresponding to different initial conditions. The initial conditions are uniformly distributed in θ and p and in the cell $-\pi \leq (p\tau)/I \leq \pi$. Show that the rotor energy, averaged over the different runs, evolves diffusively over time, i.e. linearly with the total number of kicks received since $t = 0$. What is the diffusion constant?

12. In the atmosphere of planet X, gases of He and He_2 reach a chemical equilibrium through the reaction

$$\mathrm{He} + \mathrm{He} \leftrightarrow \mathrm{He}_2$$

with densities n_A and n_B, respectively. As helium atoms interact very weakly with other helium atoms, He_2, with binding energy E_b, is the only known bound state of two He atoms. Determine the temperature of planet X in terms of E_b n_A, n_B, and fundamental constants. (Hint: Treat both He and He_2 as ideal gases. The canonical partition function of a single particle with zero internal energy is $Z = (V/h^3)(2\pi m k_B T)^{3/2}$, where m is the mass, V is the volume, h is Planck's constant, k_B is Boltzman's constant, and T is the temperature. Stirling's formula: $n! \approx \sqrt{2\pi n}(n/e)^n$)

University of Illinois at Urbana-Champaign, Spring 2006 Qualifying Examination

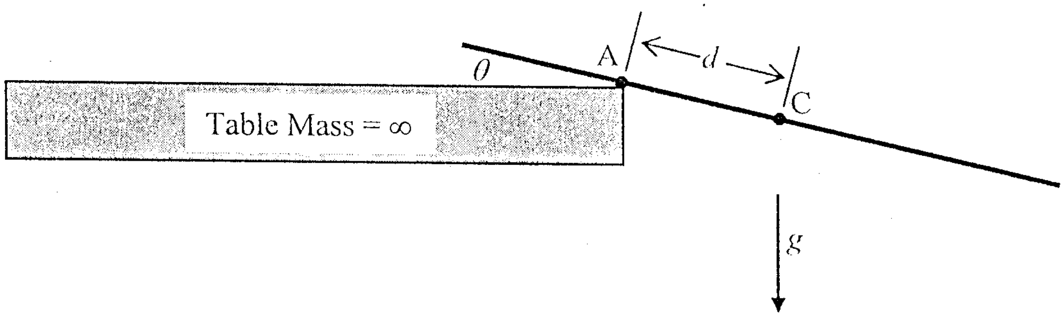
A uniform rod of mass M and length L is held flat on a horizontal table and at right angles to an edge of the table. The center of mass C of the rod projects a distance d beyond the table edge at A. The rod is released at rest. It starts to rotate about A and eventually slides off the table. The coefficient of static friction is μ.

(a) Calculate the moments of inertia of the rod, I_C about point C and I_A about point A. You may express the answers for parts (b)-(d) below in terms of I_C and I_A.

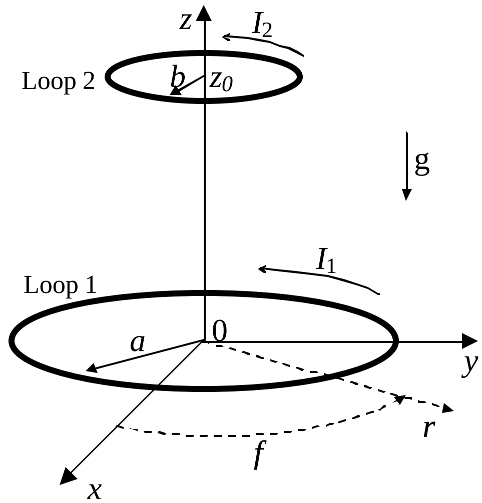
(b) Calculate the angular velocity ω of the rod as a function of the rotation angle θ before sliding occurs.

(c) The force exerted by the table edge on the rod has a component N in a direction perpendicular to the rod and the table edge. Calculate N as a function of θ before sliding occurs.

(d) Calculate the angle θ_0 when sliding begins.



A wire loop (loop 1 in the figure) of radius a, centered at the origin of a cylindrical (r, f, z) coordinate system, lies in the $z = 0$ plane and has a current $I_1 = I_0 \cos(wt)$ flowing around it. A second loop (loop 2 in the figure), of radius b, lies in the $z = z_0$ plane with its center on the z-axis. Assume that $z_0 >> a$ and $z_0 >> b$. (The figure is not to scale.) Also, ignore radiation effects.



(a) Consider the magnetic field produced at position $r = b$ and $z = z_0$ (*i.e.*, on loop 2) by the current I_1. The field is proportional to I_1:

$$B_r = a_r I_1 z_0^a$$
$$B_f = a_f I_1 z_0^b \ ,$$
$$B_z = a_z I_1 z_0^g$$

where a_r, a_f, and a_z are not functions of I_1 or z_0.

i. Show that $a_f = 0$.
ii. What are the values of a and g?

In case you could not solve part **(a)**, you may express your answers to parts **(b)**, **(c)**, and **(d)** in terms of a and g, as well as the other quantities defined in the problem.

(b) Assume that loop 2 has self inductance, L, and negligible resistance. Determine I_2, the current induced in loop 2 as a function of time. Take positive current to be in the direction of increasing f, as shown in the figure.

(c) Calculate the z-component of the force on the second loop as a function of time.

(d) Assume that w is sufficiently large that the height of loop 2 (which has mass, m) is determined only by the time averaged force. Determine the equilibrium height, z_0. Gravity, $\vec{g}$, points in the $-z$ direction.

(e) Calculate a_r and a_z in terms of a, b, z_0, and constants. Keep only the largest nonvanishing terms in a/z_0 and b/z_0.

A cesium atom used in an atomic clock is driven by a classical, oscillating magnetic field between two hyperfine states $|a\rangle$ and $|b\rangle$. The Hamiltonian which describes the atom interacting with the classical field is:

$$H = \frac{\hbar w_0}{2}\left(|a\rangle\langle a| - |b\rangle\langle b|\right) + \hbar 2\Omega \cos\left(w_0 t\right)\left(|a\rangle\langle b| + |b\rangle\langle a|\right)$$

where the "coupling" Ω is a real number.

(a) One can write the atomic wavefunction as

$$y(t) = c_a(t)|a\rangle + c_b(t)|b\rangle .$$

Using Schrödinger's equation, write down differential equations for the coefficients $c_a(t)$ and $c_b(t)$.

(b) At the beginning of an experiment, the atom is prepared in state $|a\rangle$ at time t=0. Defining the coefficients $c_a' = e^{i\frac{w_0}{2}t} c_a$ and $c_b' = e^{-i\frac{w_0}{2}t} c_b$, show that the coefficients c_a' and c_b' satisfy the simultaneous differential equations

$$i\dot{c}_a' = \Omega c_b'$$
$$i\dot{c}_b' = \Omega c_a'$$

Work in the $\Omega \ll w_0$ limit and ignore oscillations in the amplitudes $c_a(t)$ and $c_b(t)$ at frequencies much higher than Ω. Hence calculate $c_a(t)$ and $c_b(t)$.

(c) After how much time is the atom 100% likely to be measured in state $|b\rangle$?

(d) If the phase of the classical field is changed so that the Hamiltonian is

$$H = \frac{\hbar w_0}{2}\left(|a\rangle\langle a| - |b\rangle\langle b|\right) + \hbar 2\Omega \cos\left(w_0 t + f\right)\left(|a\rangle\langle b| + |b\rangle\langle a|\right)$$

and the experiment is repeated, does your answer to part (c) change? Why or why not?

(a) Consider a spinless quantum mechanical particle of mass m in a simple harmonic oscillator potential in one dimension of frequency ω, and at temperature T. The energy of the eigenstate $|n\rangle$ is $E_n = \hbar\omega(n + 1/2)$. Calculate the **partition function**, Z_1, of the particle in the oscillator, and from this result calculate its **free energy**, its **entropy**, and its **mean energy** as functions of the temperature.

(b) Consider two identical spinless Bose particles put in such an oscillator.

(i) What are the possible energy eigenstates and energies of the two bosons in the oscillator. Express the answer in terms of integers, n_1 and n_2, specifying the energy eigenstates of single particles [part (a)].

Determine the degeneracy of each of the lowest four energy levels.

(ii) Calculate the partition function of the two particles at temperature T.

(c) Consider two identical Fermi particles, of spin 1/2, put in such an oscillator, one with spin up and the other with spin down. What is the partition function of the two particles at temperature T?

(d) Now suppose that the two fermions are both in spin-up states in the oscillator.

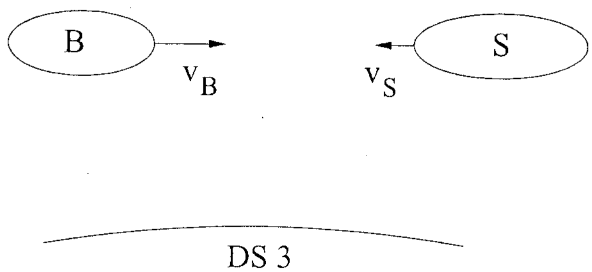
(i) What are the possible energy eigenstates and energies of the two fermions in the oscillator. Express the answer in terms of integers, n_1 and n_2, specifying the energy eigenstates of single particles. Determine the degeneracy of the lowest four energy levels.

(ii) Calculate the partition function of the two fermions in the oscillator.

(e) Compare the entropy of the two fermions in part (d) at temperature T to that of the two bosons in part (b) at the same temperature. [Hint: You could extract the entropy difference of the two systems from the ratio of their partition functions.] Explain your result in terms of the enumeration of the energy eigenstates in parts (b-i) and (d-i).

Astronauts Sally and Bob decide to settle one of their disagreements in a "paint duel". They fly their space shuttles on a collision course toward each other while keeping their shuttles at constant speed. Each shuttle is equipped with paint guns which are capable of emitting spray paint at high speed. Whoever gets the most paint on their shuttle loses. A worker from space station DS 3 is there to observe and referee the duel.

Note: Express times in units of s (seconds), velocities in units of c (speed of light), and distances in units of *light seconds*= cs.



In the frame of the space station the following events take place:

event 1: Sally fires her paint gun at $x = 9\,cs$ and $t = -2\,s$.
event 2: Bob fires his paint gun at $x = 0\,cs$ and $t = 0\,s$.
event 3: Bob's ship is hit by paint at $x = 4\,cs$ and $t = 5\,s$.
event 4: Sally's ship is hit by paint at $x = 5\,cs$ and $t = 6\,s$.

(a) Show that Sally's velocity in the rest frame of the space station is $v_S = -1/2\,c$ while Bob's velocity is $v_B = +4/5\,c$.

(b) Find the intervals $\Delta x'$ and $\Delta t'$ between events 1 and 2 in Sally's rest frame.

(c) Determine if it is possible for events 1 and 2 to be causally connected.

(d) Determine the velocity of the paint from Bob's paint gun in Bob's rest frame.

(e) Bob is trying to evade the rules of the paint duel, and sends a message on his communicator to a friend's shuttle hidden behind Sally's shuttle. Bob's communicator is using a frequency of $f = 10^{10}$Hz. Sally's receiver is able to receive communications in the frequency range of 3×10^9 to 3×10^{10}Hz. Determine whether Sally is able to receive Bob's message.

An insulating sphere of radius R contains a fixed spherically symmetric distribution of electric charge with density $r(r)$. The dependence of $r(r)$ on radius r is not specified. The sphere rotates rigidly with angular frequency vector $\mathbf{O} = \Omega\hat{z},\ \ \Omega > 0$.

(a) Give an expression for the electrostatic potential $f(\mathbf{r})$ at any point $\mathbf{r}$ in terms of an integral involving $r(r)$. The potential is zero at infinity.

(b) Give an expression for the current density vector $\mathbf{j}(\mathbf{r})$ in terms of the quantities given, at any point inside the sphere.

(c) Give an expression for the magnetic field vector $\mathbf{B(r)}$ at any point $\mathbf{r}$ in space in terms of an integral over the current density found in part (b).

(d) Find a simple expression for the ratio $\dfrac{|\mathbf{B}(0)|}{\phi(0)}$ in terms only of W and fundamental constants.

(e) Give the direction of $\mathbf{B}(0)$ if the charge density is positive everywhere.

HINT: You may find it convenient to use the vector relation

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a}\cdot\mathbf{c})\mathbf{b} - (\mathbf{a}\cdot\mathbf{b})\,\mathbf{c}$$

The helium atom consists of a nucleus with charge +2e and no spin, and two electrons, each with charge –e and spin ½$\hbar$. In this problem, treat the nucleus as a fixed (immovable) source of electric field. Also, ignore the spin-orbit interaction and other fine structure and hyperfine effects.

(a) Write the Schrödinger equation that describes this problem.

In the following two parts, ignore the interaction between the two electrons.

(b) Write a normalized 2-electron state, including the spin and spatial degrees of freedom, that has the minimum (ground state) energy. How many states are degenerate with it?

(c) What are the energies, in electron-Volts (eV), of the 2-electron ground state, E_g, and first excited state, E_2?

Now, take into account the Coulomb interaction between the electrons.

(d) The integrals needed to calculate the electron-electron interaction energy, E_{ee}, are too tedious for a qual problem. **Do not evaluate any integrals.** Inspection of the integrals will reveal that the dependence of E_{ee} on the nuclear charge, Z, is $E_{ee} \propto Z^g$ (assuming that the wave functions of the two electrons are not distorted by the e-e interaction). What is g ?

(e) Assume that you've done the integral in part **(d)** for $Z = 2$, obtaining $E_{\mathrm{ee}} = b|E_{\mathrm{g}}|$ (b is a positive number). Set up and solve a variational calculation of the 2-electron ground state energy. The variation is to let the wave function of each of the two electrons be that of an atom of some other (not necessarily integer) Z. Call it Z'. The two electrons are described by the same Z'. Determine the value of Z' that minimizes the helium ground state energy.

NOTE: The ground state wave function of an electron in the hydrogen atom is:

$$y_{100}(r) = \left(\frac{1}{pa^3}\right)^{1/2} e^{-r/a},$$

where $a = \hbar^2/me^2$, and the ground state energy is $E_1 = -me^4/2\hbar^2 = -13.6$ eV.

Consider a rigid, nonlinear molecule, such as methane, whose moment of inertia tensor is isotropic, i.e. $I_{xx} = I_{yy} = I_{zz} = I$, with all other components being zero. The rotational energy eigenvalues are $J(J+1)h^2/2I$ for integer values of J. Since this is <u>not</u> a linear diatomic molecule, the degeneracy of the J^{th} level in this case can be taken to be $(2J+1)^2$. The molecule is in thermal equilibrium at temperature T. Throughout this problem, consider only rotational degrees of freedom, ignoring translations and vibrations.

(a) State the equipartition theorem, and give an inequality for it to apply to this problem.

When the equipartition theorem is applicable, you may use it rather than doing an explicit calculation in the remainder of this problem.

Parts (b)-(d) concern *the high T limiting behavior*, with k_BT *large* compared to any characteristic energies of the system.

Give the dominant contribution to:

(b) the average rotational energy of the molecule, $E_R(T)$.

(c) the heat capacity C(T).

(d) the entropy σ (T).

In parts (e)-(g), assume that k_BT *is small* compared to any characteristic energies of the system.

Give the dominant contribution to:

(e) $E_R(T)$.

(f) C(T).

(g) σ (T).